Week 7: Report Writing

NPTEL Soft Skills · Prof. Binod Mishra · Lectures 31–35

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Lecture 31: Report Writing

Week 7 · Course book PDF pages 504–517 (printed pages 503–516)

Learning objectives

- State the origin and definition of a report.
- List the attributes of a report: formal, factual, specific purpose and audience, organised, unified, illustrated, bound.
- Explain the objectives of reports.
- Differentiate report writing from literary writing.

Important definitions

REPORT

A formal communication written for a specific purpose and audience; it describes the procedures followed for collecting and analysing data and gives conclusions and, if required, recommendations.

ETYMOLOGY OF REPORT

From Latin *reportare* = *re* (back) + *portare* (to carry): to carry information to a person who is not present. (The transcript glosses ‘re’ as ‘to send’.)

UNITY AND COHERENCE

Every sentence and paragraph links to the topic and to one another, giving a smooth flow of thought.

CROSS-FERTILISATION OF IDEAS

Exchange of ideas across people and departments through reports.

Core concepts

Why reports matter

- Decisions (promotion, product launch, new policy) rest on reports; a committee studies data and concludes.
- Doctor's medical tests are an everyday analogy of procedure, data, interpretation and conclusion.
- A report is the "bread and butter" of an organisation; it can make or mar the fortune of both organisation and individual.

Attributes of a report

1. **Formal** piece of writing.
2. **Factual** – data-based, objective, no wrong data.
3. Written for a **specific audience** (language adapted; computer professional versus layman).
4. Written for a **specific purpose**; only when required (need-based).
5. **Systematic and well organised** (beginning, discussion, end).
6. **Unity and coherence**.
7. Uses **illustrations** (charts, graphs, pie charts, tables).
8. **Bound properly** with a cover page and title; sequence of parts.

Objectives of a report

- **Update progress** of the organisation.
- **Evaluate** performance (e.g. five-year data and trends).
- **Guide the organisation to a corrective path** when things go wrong (committee, data, conclusions).
- **Spread information** internally and externally to stakeholders and shareholders.
- **Cross-fertilise ideas** and reveal gaps in reasoning.
- **Record findings and recommend action.**
- **Create awareness** and self-awareness (compare yourself with a promoted colleague).

Professional's only tangible product

A report is "a professional's only tangible product": it often is the only contact between an engineer/professional and management, so it shows reasoning, scientific attitude, grasp, intelligence and initiative; clarity is the hallmark.

Types, classifications and characteristics

Attributes of a report

Attribute	Meaning
Formal	Official style and structure
Factual	Based on accurate data
Audience-specific	Language fits reader
Purpose-specific	Written on demand for a reason
Organised	Fixed structure and sequence
Unified and coherent	Smooth flow of ideas
Illustrated	Charts, tables, graphs
Bound	Cover page, title, order

Examples and applications

EXAMPLE

Medical test reports leading to diagnosis.

EXAMPLE

Committee investigating a fire loss.

EXAMPLE

Comparing yourself with a promoted colleague.

Comparison tables

Report writing versus literary writing

Point	Report	Literary
Initiative	Written on demand	Own initiative
Audience	Specific	General, not fixed
Structure	Fixed	No fixed structure
Approach	Objective	Subjective
Language	Plain, matter-of-fact	Figurative, ornamental
Aim of reader	Information	Pleasure
Formality	Formal	Spontaneous

Important facts for MCQ revision

EXAM TIP

- Report derives from Latin reportare (re + portare).
- A report is a formal communication for a specific purpose.
- Reports are need-based, not self-initiated.
- A report is a professional's only tangible product.
- Reports contain data, procedure, conclusions and recommendations.
- Illustrations include charts, graphs, pie charts and tables.
- Objectives: update, evaluate, guide, spread information, cross-fertilise ideas, record and recommend, create awareness.
- Clarity is the hallmark of business communication.

QUIZ-RELATED CONCEPTS (QUIZ PDF PAGE REFERENCES)

Question	Quiz PDF page	Concept tested	Type
W7 Q5	p.25	Technical vs literary writing	True/False
W7 Q15	p.27	Technical reports vs literary writing	MSQ

Full question text and answers: Quiz analysis.

Summary

A report (from Latin *reportare*) is a formal, factual, systematic communication prepared on demand for a specific audience and purpose. It describes procedure and data, draws conclusions and may recommend action. Reports update, evaluate, correct, inform and create awareness. Unlike literary writing, reports are objective, structured, plain and audience-specific.

Practice questions

Q1 (MCQ) The word 'report' derives from the Latin:

A. Reportare

B. Reportum

C. Reporter

D. Rapere

> **Answer:** A
> **Why:** Re + portare.

Q2 (MCQ) Which is a feature of report language?

A. Ornamental

B. Plain and factual

C. Highly figurative

D. Exaggerated

> **Answer:** B
> **Why:** Objective and matter-of-fact.

Q3 (MSQ) Select attributes of a report.

A. Formal

B. Factual

C. Spontaneous

D. Well organised

> **Answer:** A, B, D
> **Why:** Spontaneity belongs to literary writing.

Q4 (T/F) A report is written whenever the writer feels like it.

> **Answer:** False
> **Why:** It is written on demand.

Q5 (Fill) A report is a professional's only _____ product.

> **Answer:** tangible
> **Why:** It shows his abilities.

Q6 (Short) State two objectives of a report.

> **Answer:** To update the progress of an organisation and to guide it to a corrective path (also evaluate, inform, recommend).
> **Why:** Reports support decisions.

Revision checklist

☐ I know the etymology and definition.

☐ I can list the attributes.

☐ I can list the objectives.

☐ I can differentiate report from literary writing.

☐ I recall 'tangible product'.

Lecture 32: Types of Reports

Week 7 · Course book PDF pages 518–528 (printed pages 517–527)

Learning objectives

- Classify reports by length, form, nature, frequency and purpose.
- Differentiate short (informal) and technical (formal) reports.
- Differentiate informative and analytical (interpretive) reports.
- Explain routine, management and special reports.
- Recall the bathing-suit quotation by Sherring and Christian.

Important definitions

SHORT REPORT

An informal, brief report (one page to about 10 pages) with an authorisation, a problem, a conclusion and, only if asked, a recommendation; includes letter reports and memo reports.

TECHNICAL REPORT

A formal, documented report with prefatory material, deep and systematic discussion, data analysis, illustrations and back matter.

INFORMATIVE REPORT

A report that presents information as it is, without analysis or recommendation (e.g. inspection, lab, attendance reports).

ANALYTICAL (INTERPRETIVE) REPORT

A report that interprets, analyses and evaluates data, reaches conclusions and may recommend action.

ROUTINE REPORT

A report written regularly at fixed intervals on a fixed proforma, dealing with regular matters (finance, production, sales, inventory).

MANAGEMENT REPORT

A non-technical, short report for top managers, with restricted language, journalistic tone and more illustrations.

SPECIAL REPORT

A report written during a critical juncture or aberration, with no standard format and time pressure.

ACR

Annual Confidential Report: a routine report evaluating an employee's performance.

APPENDIX

Back matter where supporting material that would block the flow of the report is placed.

Core concepts

Ways of classifying reports

- By **length**: short and long.
- By **form**: written and oral.
- By **subject / nature**: management and technical.
- By **frequency**: routine and special.
- By **purpose / function**: informative and analytical (interpretive).

Short reports

- Informal; e.g. a school picnic report; one page to 5–6 or 8–10 pages.
- Needs a **letter of authorisation** and a statement of the problem; then conclusion.
- **Recommendation only if asked:** conclusions rest on data, recommendations bring in your own thoughts.
- Letter reports (salutation, complimentary close, informal tone) and memo reports (no salutation or close) are short reports.
- Be concise, plain, direct; ask 'why am I writing this?'; review before sending; move blocking material to the appendix.

Technical reports

- Proper documentation: beginning, prefatory material, discussion, analysis, illustrations, supporting material.
- Very detailed, logical, systematic discussion to analyse a problem and provide a solution.
- **Data must be factual and its sources authentic**: wrong data leads to wrong decisions that harm the institution.
- Data is collected, analysed, interpreted and sampled to reach a conclusion (and recommendation, if required).
- Analysis is divided into sections and subsections (e.g. pollution, garbage disposal) and must lead logically to the conclusion.
- Supporting material (sources, methods) goes to the back matter.
- Divided into **informative** and **analytical**.

Informative reports

- Simply convey information as it is, not as it should be; no analysis, usually shorter.
- Often a fixed proforma with ticks; little language use; the writer does not recommend action.
- Examples: market survey, laboratory report, inventory, inspection, traffic observation.

- Also: periodic (weekly attendance, monthly inspection), readership surveys, employee booklets, prospectus, website information, committee reports, policy statements, sales, credit and production reports.

Analytical (interpretive) reports

- Interpret, analyse and evaluate data; investigate a problem (e.g. a product with deficiencies) and may recommend action.
- Writer must be **unbiased** and keep personal bias out.
- Elaborate and descriptive: may run 30–35 pages.
- Example: a Pay Commission examines situations, recommends, and government decides.
- More objective and scientific; analysis must serve the common good.
- Busy readers read the illustrations (tables, charts, graphs), so use relevant visuals.

Routine reports

- Regular, at fixed intervals (fortnight, month, year); fixed proforma; little language.
- Deal with finance, production, sales, performance, inventory; keep internal people and outside agencies informed.
- Examples: annual reports, project progress proforma (every six months), lab report, stock verification, inspection reports (e.g. car makers withdrawing or adding features), ACR (annual confidential report).

Management reports

- **Non-technical**, short, for top people; more illustrations, little explanation.
- Language restricted, tone at times journalistic.
- Managers read the conclusion and may refer to the technical report for detail.
- Writer must know the readers' background, age, taste and preferences.

Special reports

- Written only in critical hours or after a mishap; low frequency; time limit.
- No standard format; writer relies on caution and felicity of language.
- Writer must be extra vigilant, diligent, intelligent and careful because action will follow.

Key quotation

Sherring and Christian: "A report is like a bathing suit; it covers everything that has to be covered, but nothing more." (No unnecessary or irrelevant material.)

Types, classifications and characteristics

Classification of reports

Basis	Types
Length	Short, long
Form	Written, oral
Subject	Management, technical
Frequency	Routine, special
Function	Informative, analytical (interpretive)

Examples of informative and routine reports

Category	Examples
Informative	Lab report, inventory, inspection, market survey, prospectus, committee report, sales, credit, production report
Routine	Annual report, progress proforma, stock verification, inspection report, ACR

Theories, principles, models and contributors

BATHING-SUIT QUOTATION — SHERRING AND CHRISTIAN

A report covers everything that has to be covered, but nothing more.

Examples and applications

EXAMPLE

School picnic report (short report).

EXAMPLE

Report on pollution or garbage disposal (technical).

EXAMPLE

Teacher’s attendance record deciding course continuation.

EXAMPLE

Pay Commission recommendations to the government.

EXAMPLE

Car companies withdrawing features after inspection reports.

Comparison tables

Informative versus analytical report

Point	Informative	Analytical (interpretive)
Content	Information as it is	Interpretation and evaluation
Recommendation	Usually none	Often given
Length	Shorter	Longer (30–35 pages)
Language	Little; proforma ticks	Elaborate, descriptive
Use	Record and trend	Solve problems, take decisions

Short versus technical report

Point	Short	Technical
Formality	Informal	Formal
Length	1–10 pages	Long
Illustrations	Few	Many
Structure	Authorisation, problem, conclusion	Prefatory, body, back matter
Examples	Letter and memo reports	Pollution report

Routine versus special report

Point	Routine	Special
Frequency	Regular intervals	Rare, on critical occasions
Format	Fixed proforma	No standard format
Language use	Little	Careful, meticulous
Time	Scheduled	Limited, urgent

Important facts for MCQ revision

EXAM TIP

- Reports are classified by length, form, subject, frequency and function.
- Letter and memo reports are short reports.
- Do not recommend unless asked; conclusion is fine.
- Technical report needs authentic data and sources.
- Technical reports are informative or analytical.
- Analytical reports can run 30–35 pages.
- Routine reports follow fixed proformas at fixed intervals.
- ACR = Annual Confidential Report.
- Management reports are non-technical, short, illustrated.
- Special reports have no standard format and are rare.
- Bathing-suit quote by Sherring and Christian.

QUIZ-RELATED CONCEPTS (QUIZ PDF PAGE REFERENCES)

Question	Quiz PDF page	Concept tested	Type
W7 Q7	p.26	Types of reports	MCQ

Full question text and answers: Quiz analysis.

Summary

Reports vary by length (short or long), form (written or oral), subject (management or technical), frequency (routine or special) and function (informative or analytical). Short reports are informal; technical reports are formal and documented and must rest on authentic data. Informative reports present facts; analytical reports interpret and may recommend. Routine reports are periodic and proforma-based, management reports are short and non-technical, and special reports arise in critical situations. A report should cover everything necessary, but nothing more.

Practice questions

Q1 (MCQ) A letter report is a type of:

A. Long report

B. Short report

C. Special report

D. Oral report

> **Answer:** B
> **Why:** Letter and memo reports are short reports.

Q2 (MCQ) Which report simply presents information as it is?

A. Analytical

B. Informative

C. Special

D. Management

> **Answer:** B
> **Why:** No analysis or recommendation.

Q3 (MCQ) The ACR is an example of a:

A. Special report

B. Routine report

C. Oral report

D. Management report

> **Answer:** B
> **Why:** Annual and periodic.

Q4 (MSQ) Which are features of management reports?

A. Non-technical

B. Short

C. Journalistic tone possible

D. Heavy documentation

> **Answer:** A, B, C
> **Why:** Heavy documentation is technical.

Q5 (T/F) A report writer should always give recommendations.

> **Answer:** False
> **Why:** Only if asked.

Q6 (Fill) A report is like a _____ suit: it covers everything necessary but nothing more.

> **Answer:** bathing
> **Why:** Sherring and Christian.

Q7 (Short) Distinguish informative and analytical reports.

> **Answer:** Informative reports present facts as they are; analytical reports interpret and evaluate data and may recommend action.
> **Why:** Core classification.

Revision checklist

☐ I can classify reports by five bases.

☐ I know short vs technical.
☐ I know informative vs analytical.
☐ I can describe routine, management and special reports.
☐ I recall the bathing-suit quote.

Lecture 33: Strategies for Report Writing Part I

Week 7 · Course book PDF pages 529–545 (printed pages 528–544)

Learning objectives

- List the steps in preparing to write a report: purpose and scope, audience, data collection, organising material, outline.
- Explain terms of reference (TOR) and why knowing the reader matters.
- Distinguish primary and secondary sources of data.
- Evaluate personal observation: process, precautions, limitations.
- State the advantages and disadvantages of telephonic interviews.

Important definitions

TERMS OF REFERENCE (TOR)

Instructions from the person assigning the report stating the problem, scope, readers and what is to be done.

SCOPE

The extent of the problem the report covers and who will read it.

PRIMARY SOURCE

First-hand data from personal observation and investigation.

SECONDARY SOURCE

Existing data used to verify or supplement: library, Internet, internal records.

PERSONAL OBSERVATION

Seeing with a purpose; the word is explained in the lecture as ob + servation.

CONTROLLED OBSERVATION

Observation where the subject knows or is prepared in advance.

UNCONTROLLED OBSERVATION

Observation where the subject is unaware of being observed.

PRIMARY READER

The boss or person who assigned the report.

SECONDARY READER

Other readers, including those outside the organisation with relevant expertise.

TELEPHONIC INTERVIEW

Collecting data by planned telephone questioning.

Core concepts

Steps before writing a report

1. Identify **purpose and scope**.
2. Determine the **audience**.
3. Choose the right **method of data collection**.
4. **Organise** the material.
5. Make an **outline**.

Purpose, scope and TOR

- Ask: why am I writing, and who gave me the task?
- The assigner supplies the **terms of reference**: problem, scope, readers, what to cover.
- Example: Mess Secretary asked by the Chief Warden to report on mess functioning (timing, food quality, vegetarian meals, sweets).

Knowing the audience

- Readers: superior officers, colleagues, counterparts, subordinates, people of other organisations, shareholders, customers, the public. (In the example: Chief Warden, Dean Students' Welfare, Deputy Director, Director.)
- Know their background, your relationship with them and their **attitude** (favourable or hostile); this guides language and presentation.
- Primary reader = boss; secondary readers = others, including outsiders with expertise.
- Do not be influenced by the assigner's expectations: work on the data to stay unbiased.

Sources of data

- **Primary:** personal observation and investigation (verified later).
- **Secondary:** library, Internet, internal records.
- Ways of collection: personal observation, telephonic interviews, personal interviews, questionnaires.
- Use specific words in search engines; avoid straying to irrelevant sites; verify authenticity with other sources without wasting time.

Personal observation: how to do it

- Fix beforehand what data you want (a habit, trend, quality).
- Decide between **controlled** and **uncontrolled** observation; people change behaviour when they know they are watched.
- Do not display pen and paper; note afterwards from memory when observing people (open note-taking is fine for traffic counts).

- Ignore sudden irrelevant changes (posture, gesture); stay focused.
- Avoid prejudice and see with the 'mind's eye'; first impressions (e.g. a tired-looking speaker) may be wrong.
- Covert recording that is later disclosed creates problems; be specific, not general.

Limitations of observation

- Cannot be the only method; gives first-hand but not necessarily representative data.
- Can **report behaviour but not interpret it**.
- Biased by the observer; one day may not represent others (e.g. rainy Sunday traffic).
- Slow, costly, and subjects may be unavailable.

Telephonic interviews

Advantages: accessible anywhere, saves time and travel, wide geographic reach, low refusal rate.

Disadvantages: needs meticulous planning and specific questions; casual talk differs from data-collecting talk; detailed data is hard to get; no sight of non-verbal cues (video calls help only partly); cannot judge age or nationality or if the respondent hides information; limited time, so follow-up is hard; respondents may give wrong information or send someone else.

Types, classifications and characteristics

Sources of data

Source type	Examples
Primary	Personal observation, investigation
Secondary	Library, Internet, internal records

Methods of data collection (Part I)

Method	Key point
Personal observation	First-hand; cannot interpret; observer bias
Telephonic interview	Wide reach; limited depth
Personal interview	Discussed next lecture
Questionnaire	Discussed next lecture

Examples and applications

EXAMPLE

Mess Secretary's report for the Chief Warden.

EXAMPLE

Observing Sunday traffic flow.

EXAMPLE

Misjudging a tired-looking speaker.

EXAMPLE

Searching for air pollution data.

Comparison tables

Primary versus secondary sources

Point	Primary	Secondary
Nature	First-hand	Existing material
Methods	Observation, investigation	Library, Internet, records
Role	Starting data	Verification and support

Controlled versus uncontrolled observation

Point	Controlled	Uncontrolled
Subject aware?	Yes, in advance	No
Risk	Subject prepares or hides things	Less altered behaviour

Important facts for MCQ revision

EXAM TIP

- Preparation: purpose and scope, audience, data method, organisation, outline.
- TOR = terms of reference.
- Know the attitude of readers.
- Primary sources: observation and investigation.
- Secondary sources: library, Internet, internal records.
- Observation reports behaviour but cannot interpret it.
- Do not recommend or take sides under the boss’s influence.
- Telephonic interviews: low refusal, wide reach, limited detail.

QUIZ-RELATED CONCEPTS (QUIZ PDF PAGE REFERENCES)

Question	Quiz PDF page	Concept tested	Type
W7 Q1	p.25	Primary and secondary sources	True/False
W7 Q2	p.25	Personal observation	True/False
W7 Q3	p.25	Interview as a data-collection method	True/False
W7 Q8	p.26	Secondary sources	MCQ
W7 Q10	p.26	Personal observation	MCQ
W7 Q11	p.26	Preparatory steps for report writing	MSQ
W7 Q12	p.27	Observation: advantages and disadvantages	MSQ

Full question text and answers: Quiz analysis.

Summary

Preparing a report starts with the purpose and scope defined in the terms of reference, then analysis of the audience and its attitude, selection of data-collection methods, organisation of material and an outline. Data come from primary sources (personal observation, investigation) and secondary sources (library, Internet, internal records). Observation gives first-hand data but is biased, slow and cannot interpret behaviour; telephonic interviews reach many people cheaply but lack depth and non-verbal cues.

Practice questions

Q1 (MCQ) TOR stands for:

A. Terms of Reference
B. Time of Report
C. Type of Research
D. Tone of Reader

> **Answer:** A
> **Why:** Instructions given by the assigner.

Q2 (MCQ) Which is a secondary source?

A. Personal observation
B. Investigation
C. Library
D. Eyewitness view

> **Answer:** C
> **Why:** Library, Internet and internal records.

Q3 (MSQ) Disadvantages of telephonic interviews include:

A. No non-verbal cues

B. Limited detail

C. Wide reach

D. Respondent may mislead

> **Answer:** A, B, D
> **Why:** Wide reach is an advantage.

Q4 (T/F) Observation can both report and interpret behaviour reliably.

> **Answer:** False
> **Why:** It reports but cannot interpret.

Q5 (Fill) The boss who assigns the report is the _____ reader.

> **Answer:** primary
> **Why:** Others are secondary.

Q6 (Short) Why should note-taking be discreet during observation?

> **Answer:** People who know they are observed change behaviour, defeating the purpose.
> **Why:** Controlled versus uncontrolled observation.

Revision checklist

☐ I know the five preparation steps.

☐ I can define TOR.

☐ I can classify data sources.

☐ I know observation limits.

☐ I know telephone interview pros and cons.

Lecture 34: Strategies for Report Writing Part II

Week 7 · Course book PDF pages 546–560 (printed pages 545–559)

Learning objectives

- Define a personal interview and list its advantages and precautions.
- Explain questionnaires, their advantages and disadvantages.
- List the types of questions: closed, rating, open-ended, multiple choice.
- Apply guidelines for framing a questionnaire and covering letter.
- Recall Freud's and Ruskin's quotations.

Important definitions

PERSONAL INTERVIEW

A process of seeking information directly through a face-to-face conversation with a purpose.

QUESTIONNAIRE

A set of standardised questions sent to many respondents to collect data without travel.

CLOSED QUESTION

A question with a fixed answer such as yes/no (e.g. 'Do you smoke?').

RATING QUESTION

A question asking the respondent to rate on a scale such as 1 to 10.

OPEN-ENDED QUESTION

A question allowing respondents to express their own opinion in their own words.

MULTIPLE-CHOICE QUESTION

A question with (typically four) offered answers.

LEADING QUESTION

A question that suggests the expected answer or is embarrassing or tricky.

COVERING LETTER

A letter accompanying a questionnaire explaining its purpose and assuring confidentiality.

REPRESENTATIVE SAMPLE

A group of respondents whose answers reflect the whole population.

Core concepts

Why other methods are needed

Observation and telephonic interviews have limits and their data may not be representative, so data should be verified through personal interviews and questionnaires.

Personal interview: advantages

- Face-to-face communication is better because **non-words** (dress, bearing, gestures, postures, silence, pauses) are available.
- Gathers **attitudes and feelings**; faces reveal emotions.
- Learns from people with intimate knowledge of the problem; verifies and supplements observation, questionnaire or telephone data.
- The interviewer has **control** and can bring the interviewee back to the point.
- Questions can be interpreted and ambiguities cleared; replies are more representative (people may lie on the phone but less so in person).
- Gives first-hand impressions and immediate feedback.

Freud: "He who has eyes to see and ears to hear can convince himself that no mortal can keep a secret. If his lips are silent, he chatters with his fingertips; betrayal oozes out of him at every pore."

Personal interview: tips and disadvantages

- Costly and needs travel.
- Seek an **appointment in advance**, state the purpose; be courteous and tactful.
- Standardise questions; limit time (busy professionals).
- Jot notes if allowed; recording is possible.
- **Do not argue**; give the interviewee room; speak less, listen carefully; take a subordinate role.
- Avoid embarrassing questions that affect the interviewee's image; frame questions subtly.

Questionnaire: advantages

- Saves travel, time and money; reaches people in different locations with one set of questions.
- Convenient for respondents, who can **look up information** at leisure.
- No interviewer bias; respondents are fearless and may stay **anonymous**.
- Standardised questions; follow-up questions can follow closed ones.
- Return postage can be prepaid.

Questionnaire: disadvantages

- Low response; people decide on likes and dislikes, so the sample may not be representative.
- Uncertainty; questions left unanswered or answered inadequately or meaninglessly.
- Reluctance on confidential or complex matters.
- No control over the situation.
- Cost per mailing is low but cost per return is high.

Framing a questionnaire

- Keep it **short, specific and attractive**; too long means unanswered.
- Two parts: **personal information first**, then the main questions.
- Use simple wording; arrange logically with coherence.
- Place **tricky questions in the middle**, not at the beginning or the end.
- **Avoid leading and embarrassing questions** (e.g. about alcohol brand, newspaper read).
- Be free of grammatical errors.
- Include a **covering letter** stating purpose and assuring confidentiality.

Ask direct questions: 'Do you smoke?' rather than 'What is your reaction to smoking?'.

Closing quotation

John Ruskin: "To be able to ask a question clearly is two-thirds of the way to getting it answered."

After collection: analyse, organise, outline, then write.

Types, classifications and characteristics

Types of questions in a questionnaire

Type	Description	Example / use
Closed	Yes/no answers	Do you smoke?
Rating	Scale such as 1–10	Rate the service
Open-ended	Own opinion, space provided	Reveals attitudes and trends
Multiple choice	Choose from four options	Fixed alternatives

Methods of data collection

Method	Main strength	Main weakness
Observation	First-hand	Cannot interpret; biased
Telephonic interview	Wide reach, low refusal	Limited detail, no cues
Personal interview	Non-verbal cues, control	Costly, travel
Questionnaire	Cheap, wide, anonymous	Low and unrepresentative response

Theories, principles, models and contributors

FREUD ON BODY LANGUAGE — SIGMUND FREUD

No one can keep a secret; the body betrays emotions.

RUSKIN ON QUESTIONS — JOHN RUSKIN

A clearly asked question is two-thirds answered.

Examples and applications

EXAMPLE

Smoking research: ‘Do you smoke?’

EXAMPLE

Tricky or leading: ‘Do you drink alcohol?’, ‘Which brand?’

EXAMPLE

Pre-paid return postage to improve response.

Comparison tables

Personal interview versus questionnaire

Point	Personal interview	Questionnaire
Control	With interviewer	None
Cost / travel	High	Low
Non-verbal cues	Available	Absent
Bias	Interviewer bias possible	Eliminated
Anonymity	None	Possible
Response	High	Often low

Important facts for MCQ revision

EXAM TIP

- Personal interview = conversation with a purpose.
- Non-verbal cues are available in personal interviews.
- Do not argue in an interview; listen more.
- Seek appointments in advance.
- Questionnaires save time and travel and allow anonymity.
- Types: closed, rating, open-ended, multiple choice.
- Put tricky questions in the middle.
- Avoid leading and embarrassing questions.
- A covering letter assures confidentiality.
- Ruskin: question asked clearly is two-thirds answered.

QUIZ-RELATED CONCEPTS (QUIZ PDF PAGE REFERENCES)

Question	Quiz PDF page	Concept tested	Type
W7 Q4	p.25	Personal interview	True/False
W7 Q14	p.27	Questionnaires	MSQ

Full question text and answers: Quiz analysis.

Summary

Personal interviews give reliable, verifiable, first-hand data with non-verbal cues and interviewer control but are costly; the interviewer should plan, listen, avoid arguments and embarrassing questions. Questionnaires reach many respondents cheaply and anonymously but risk low and unrepresentative responses. A good questionnaire is short, simple, logical, error-free, uses closed, rating, open-ended and multiple-choice questions, avoids leading questions, places tricky ones in the middle and carries a covering letter. Ruskin: a clearly asked question is two-thirds answered.

Practice questions

Q1 MCQ A key advantage of a personal interview is:

A. Low cost
B. Access to non-verbal cues
C. Anonymity
D. No travel

> **Answer:** B
> **Why:** Face-to-face reveals non-words.

Q2 MCQ Where should tricky questions be placed?

A. First
B. Middle
C. Last
D. Nowhere

> **Answer:** B
> **Why:** Neither in the beginning nor at the end.

Q3 (MSQ) Which are question types for questionnaires?

A. Closed

B. Rating

C. Open-ended

D. Hostile

> **Answer:** A, B, C
> **Why:** Multiple choice is also a type.

Q4 (T/F) A questionnaire gives the researcher control over the respondent.

> **Answer:** False
> **Why:** There is no control; that is a limitation.

Q5 (Fill) A questionnaire should be accompanied by a _____ letter.

> **Answer:** covering
> **Why:** It explains purpose and assures secrecy.

Q6 (Short) Name two disadvantages of questionnaires.

> **Answer:** Low response rate and unrepresentative or inadequate answers (also unanswered questions, reluctance on confidential matters).
> **Why:** Cost per return is high.

Revision checklist

☐ I can define personal interview and questionnaire.

☐ I know tips for interviewing.

☐ I can list question types.

☐ I know questionnaire guidelines.

☐ I recall Freud and Ruskin.

Lecture 35: Evaluation and Organization of Data

Week 7 · Course book PDF pages 561–575 (printed pages 560–574)

Learning objectives

- Explain why collected data must be evaluated and organised.
- Describe probability and non-probability sampling and their types.
- Apply note-making techniques.
- State the principles of an outline: indenting, numbering, coordination, subordination, parallel construction, logical ordering.
- Prepare a decimal-numbered outline.

Important definitions

SAMPLING

Selecting a representative part of the data so that conclusions can be drawn for the whole.

PROBABILITY SAMPLING

Sampling in which every item has a known chance of selection: random, systematic or stratified.

RANDOM SAMPLING

Selecting items purely by chance so that every item has an equal chance; the ‘purest’ way.

SYSTEMATIC SAMPLING

Choosing records by a fixed system, e.g. every ninth record.

STRATIFIED SAMPLING

Dividing the data into strata (e.g. men/women, students/teachers, children/adults) and sampling from each.

NON-PROBABILITY SAMPLING

Sampling by convenience or judgment of the analyst.

OUTLINE

A rough, tentative, short structure of the whole report; a distribution table showing which information goes to which section.

COORDINATION

Main head and sub-heads of the same level are of equal importance and belong to the same subject.

SUBORDINATION

Sub-heads are less important than, and visibly lower than, the main head (indent and typeface).

PARALLEL GRAMMATICAL CONSTRUCTION

Headings of the same level use the same grammatical form (all nouns, all phrases).

NOTE CARDS

Cards with sections and sub-sections used to sort jumbled data.

Core concepts

From data to report

Collected data is huge and haphazard. Steps: **evaluate** (validate, sample) → **make notes** → **organise** → **outline** → write.

Sampling methods

- **Probability:** random, systematic (e.g. every ninth record), stratified (groups by criteria).
- **Non-probability:** convenience, judgment.
- Aim: representative data so the report can lead to a solution.

Making notes

- Notes later create the outline; report writing is a systematic process.
- Place quoted words in **quotation marks** and give credit.
- Add a small summary; use abbreviations (shorthand is abandoned now).
- Use **note cards** with sections and sub-sections.
- Write clearly so you can read your own notes.
- Record observations, facts and trends under suitable categories.

Principles of outline

1. **Indenting** – sub-topics shifted to the right, main topics at left.
2. **Numbering** – Roman, alphabetical or **decimal** (preferred: 1, 2, 3.1, 3.2.1); do not go **beyond the third level**.
3. **Coordination** – no new matter unrelated to the head.
4. **Subordination** – sub-heads visibly less important (bold head, normal sub-heads).
5. **Parallel grammatical construction** – all nouns/phrases/adjectives, not a mix.
6. **Logical ordering** – a sequence such as introduction, history, discussion, conclusion, recommendation, bibliography; stick to the pattern chosen.

7. **Phrasing** – use phrases, not full sentences.

Sample outlines

Advertising in India: 1 Introduction; 2 History; 3 Types → 3.1 Print, 3.2 Electronic → 3.2.1 (sub-type).

Modes of presentation: reading from manuscript, impromptu, memorisation (coordinated, no new matter).

Benefits of advertising: economic, political, moral, religious, cultural (all adjectives; not 'politics' or 'moralise').

Advantages of nationalisation: mobilisation of national resources, promotion of agricultural production… (nouns, not 'to promote').

Role of the outline

- A tailor's analogy: cut cloth into pieces, then stitch into one whole.
- Outline is a tentative guide that reminds you what is missing or misplaced; the writer may change it.
- "Once the outline is ready, half the job is done."
- Next: structure (anatomy) of the report.

Types, classifications and characteristics

Sampling methods

Category	Types
Probability	Random, systematic, stratified
Non-probability	Convenience, judgment

Principles of outline

Principle	Meaning
Indenting	Sub-topics to the right
Numbering	Decimal preferred; max three levels
Coordination	Same-level items belong together
Subordination	Lower items less prominent
Parallelism	Same grammatical form
Logical ordering	Systematic sequence

Examples and applications

EXAMPLE

Outline for 'Advertising in India' with decimal numbering 3.2.1.

EXAMPLE

Parallel list: economic, political, moral, religious, cultural.

EXAMPLE

Stratified groups: men/women, student/teacher, child/adult.

EXAMPLE

Tailor cutting and stitching cloth.

Comparison tables

Probability versus non-probability sampling

Point	Probability	Non-probability
Basis	Chance or system	Convenience or judgment
Types	Random, systematic, stratified	Convenience, judgment
Representativeness	Higher	Depends on analyst

Important facts for MCQ revision

EXAM TIP

- Evaluate data by sampling.
- Probability sampling: random, systematic, stratified.
- Systematic example: every ninth record.
- Non-probability: convenience, judgment.
- Quote with quotation marks.
- Decimal numbering preferred.
- Do not go beyond three levels.
- Parallel grammatical construction in headings.
- Outline uses phrases, not sentences.
- Outline is a tentative guide: half the job done.

QUIZ-RELATED CONCEPTS (QUIZ PDF PAGE REFERENCES)

Question	Quiz PDF page	Concept tested	Type
W7 Q6	p.25	Sampling methods	MCQ
W7 Q11	p.26	Preparatory steps for report writing	MSQ

Full question text and answers: Quiz analysis.

Summary

Collected data must be evaluated through sampling (probability: random, systematic, stratified; non-probability: convenience, judgment), noted carefully with quotation marks, abbreviations and note cards, and organised into an outline. The outline follows the principles of indenting, numbering (decimal, up to three levels), coordination, subordination, parallel grammatical construction and logical ordering, uses phrases rather than sentences and serves as a tentative guide; a good outline means half the report is done.

Practice questions

Q1 (MCQ) Which is a probability sampling method?

A. Convenience

B. Judgment

C. Stratified

D. Casual

> **Answer:** C
> **Why:** Random, systematic and stratified.

Q2 (MCQ) The preferred numbering system for outlines is:

A. Roman

B. Alphabetical

C. Decimal

D. Mixed

> **Answer:** C
> **Why:** It avoids confusion.

Q3 (MSQ) Principles of outline include:

A. Coordination

B. Subordination

C. Parallelism

D. Rhyme

> **Answer:** A, B, C
> **Why:** Rhyme is irrelevant.

Q4 (T/F) An outline should be written in full sentences.

> **Answer:** False
> **Why:** Phrases are used.

Q5 Fill Divisions in an outline should not go beyond the _____ level.

> **Answer:** third
> **Why:** Else it becomes sketchy.

Q6 Short What is parallel grammatical construction? Give an example.

> **Answer:** Using the same grammatical form for headings of one level, e.g. economic, political, moral, religious, cultural (all adjectives).
> **Why:** Avoid mixing 'politics' or 'moralise'.

Revision checklist

☐ I know sampling types.
☐ I can make notes properly.
☐ I can list outline principles.
☐ I can draft a decimal outline.
☐ I know parallelism.